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```
clear all; close all; clc;
```

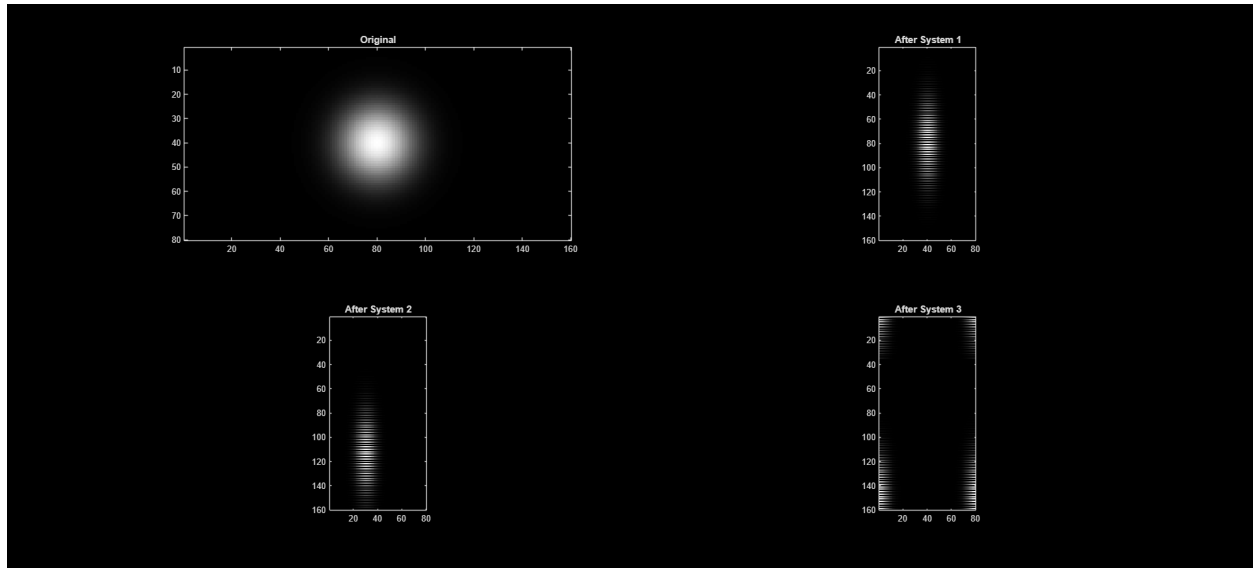
Lab 6, Images in Greyscale and RGB.

Exercise 6.1

```
% USER DEFINED VARIABLES
w = 15; % Width
x = 1:160; % Horizontal Axis
y = 1:80; % Vertical Axis
% == > Our grayscale value, x and y define location, z the value, centered
% (80, 40) <==
z = round (127* exp ( -1/ w .^2*(( y .' -40) .^2+( x -80) .^2) ) ) ;
% == > Lets MATLAB know we are working in grayscale <==
colormap ( gray ) ;
% == > Downsamples half of our x values, and upsamples our y axis <==
[ xs , ys , zs ] = image_system1 ( z ,2 ,2) ;
za = image_system2 ( zs , -10 ,35) ;
zb = image_system3 ( za , -30 ,35) ;

% PLOT RESULT WITH SUBPLOT
figure (1) ;
subplot (2 ,2 ,1) ; % == > Create 2x2 plot grid, place in top left <==
imagesc ( x , y , z ) ; % == > Display image with x y position and z value <==
axis image ; % == > 1:1 aspect ratio for image to display clearly <==
title ( ' Original ' )
subplot (2 ,2 ,2) ; % == > Places in top right of 2x2 grid <==
imagesc ( xs , ys , zs ) ; % == > Display image with xs ys position and zs
value <==
axis image ; % == > 1:1 aspect ratio for image to display clearly <==
title ( ' After System 1 ' )

subplot (2 ,2 ,3) ; % == > Places in bottom left of 2x2 grid <==
imagesc ( xs , ys , za ) ; % == > Display image in xs ys position and za
value <==
axis image ; % == > 1:1 aspect ratio for image to display clearly <==
title ( ' After System 2 ' )
subplot (2 ,2 ,4) ; % == > Places in bottom right of 2x2 grid <==
imagesc ( xs , ys , zb ) ; % == > Display image in xs ys position and zb
value <==
axis image ; % == > 1:1 aspect ratio for image to display clearly <==
title ( ' After System 3 ' )
```



Exercise 6.2

```
lighthouse = load('lighthouse.mat');
data = lighthouse.xx;

[xs, ys, zf] = image_sample(data, 2);

figure;
imagesc(xs, ys, zf);
colormap(gray);
axis image;
title('Lighthouse Sampled');

% Part D/E

data_aa = image_antialias(data);
data_aax6 = data;

for i = 1:6

    data_aax6 = image_antialias(data_aax6);

end

% Part E/F

[xz, yz, lighthouse_zeros] = image_insertzeros(data_aa, 2);

lighthouse_interpolated = lighthouse_zeros;
for i = 1:6
    lighthouse_interpolated = image_antialias(lighthouse_interpolated);
end

% Part D
```

```

figure;
subplot(1,4,1);
imshow(data);
title('Original');
subplot(1,4,2);
imagesc(xs, ys, zf);
colormap(gray);
axis image;
title('Sampled');
subplot(1,4,3);
imshow(data_aa);
title('Anti Aliased (1x)');
subplot(1,4,4);
imshow(data_aax6);
title('Anti Aliased (6x)');

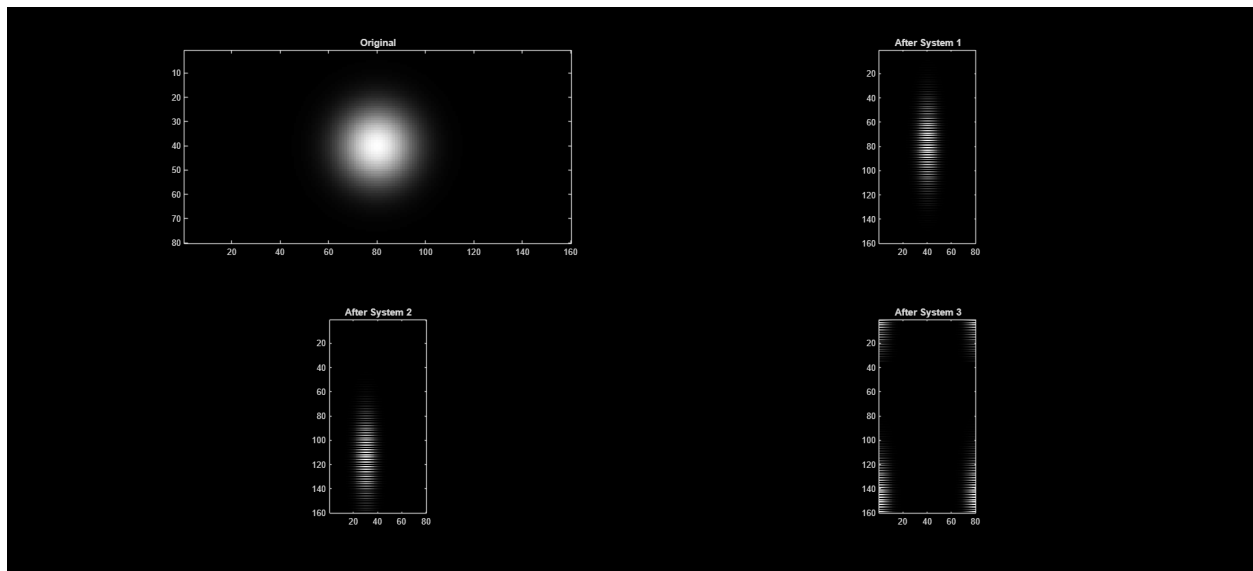
% Antialiasing smooths edges and causes blur, good for removing sharp
% edges, and functions by averaging the weight of the pixel sampled with
% pixels in immediate contact. Useful for rendering games to reduce jagged
% edges.

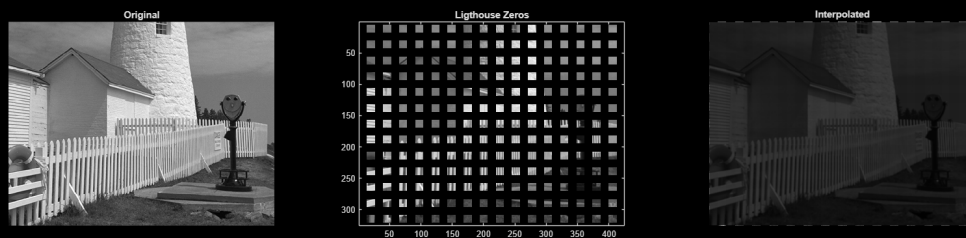
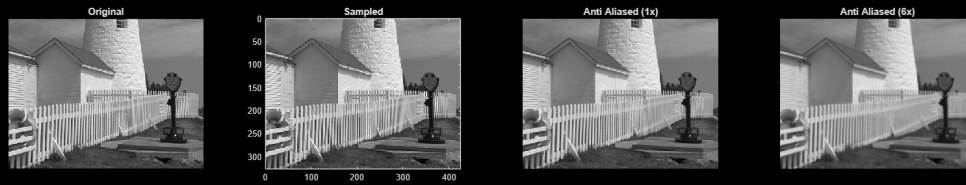
% Part F

figure;
subplot(1,3,1);
imshow(data);
title('Original');
subplot(1,3,2);
imagesc(xs, ys, lighthouse_zeros);
colormap(gray);
axis image;
title('Lighthouse Zeros');
subplot(1,3,3);
imshow(lighthouse_interpolated);
title('Interpolated');

% The zeros function causes a grid pattern over the image to be removed,
% and interpolation refills this in by averaging the values of nearby
% pixels, causing the smoothest look, although it does cause the saturation
% range to shrink, which can be beneficial if you want a homogenous look,
% but might cause issues when the difference is important.

```





Functions

```
function [ xs , ys , zs ] = image_system1 ( z , Dx , Uy )

    % IMAGE_SYSTEM1 === > Takes original z matrix and upsamples/downsamples
    <===
    % == > Pre-allocates memory to the new down/upsampled matrix <==
    zs = zeros ( ceil ( Uy * size ( z ,1) ) , ceil ( size ( z ,2) / Dx ) ) ;
    % == > Uy upsamples by taking Uy times as many row values and Dx
downsamples
    % by dividing the amount of samples by Dx <==
    ys = 1: ceil ( Uy * size ( z ,1) ) ;
    xs = 1: ceil ( size ( z ,2) / Dx ) ;
    % == > Takes every Uy rows and takes a column every Dx columns <==
    zs (1: Uy : end ,1: end ) = z (1: end ,1: Dx : end ) ;
```

```

end

function [ za ] = image_system2 ( z , Sx , Sy )
    % IMAGE_SYSTEM2 == > Transforms the image, shifting it based on Sx, Sy
    <===
    % ===== > Creates a 2d matrix with the same rows and column data as z
    <=====
    za = zeros ( size ( z ,1) , size ( z ,2) ) ;
    for nn = 1: size ( z ,1)
        for mm = 1: size ( z ,2)
            % ===== > Checks if nn - Sy and mm - Sx are within the image bounds
            when transformed <=====
                if nn > Sy && nn - Sy < size ( z ,1) && mm > Sx && mm - Sx < size
(z ,2)
                    % ===== > Lowers z value, dimming, and transforms the image
                    <=====
                        za ( nn , mm ) = 1/2* z ( nn - Sy , mm - Sx ) ;
                    end
                end
            end
        end
    end

function [ zb ] = image_system3 ( za , Sx , Sy )

    % IMAGE_SYSTEM3 == > Causes the image to invert, wrapping it around the
    edge of the frame <===

    % ===== > Recreates the matrix with one row and column removed <=====
    x = 0:1: size ( za ,2) -1;
    y = 0:1: size ( za ,1) -1;

    % ===== > Uses modulus to shift and wrap around our rows and columns <=====
    xs = mod ( x - Sx , size ( za ,2) ) ;
    ys = mod ( y - Sy , size ( za ,1) ) ;

    % ===== > Adds one so we dont have zb(0) and recreates matrix with
    wraparound <=====
    zb = za ( ys +1 , xs +1) ;

end

function [xs, ys, zs] = image_sample(z, D)

    ys = 1:D:size(z,1);
    xs = 1:D:size(z,2);
    zs = z(ys, xs);

end

function zaa = image_antialias(z)

    [M, N] = size(z);
    zaa = z;

```

```

        for y = 2:M-1
            for x = 2:N-1
                zaa(y, x) = 0.5*z(y, x) + 0.125*(z(y-1, x) + z(y+1, x) + z(y,
x-1) + z(y, x+1));
            end
        end

end

function [ xz , yz , zz ] = image_insertzeros (zaas , U)

    [M, N] = size(zaas);
    Mz = M + (M-1)*(U-1);
    Nz = N + (N-1)*(U-1);
    zz = zeros(Mz, Nz);
    zz(1:U:end, 1:U:end) = zaas;
    xz = 1:Nz;
    yz = 1:Mz;

end

```

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